

How Do Strategy and Leadership Styles Jointly Affect Co-development and Its Innovation Outcomes?*

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Co-development with customers attracts considerable interest as a means to improve companies' product innovativeness and performance. Forward-thinking companies integrate co-development within their business models, but many remain uncertain about which levers best foster its implementation. This study therefore takes a top-down perspective on ways to stimulate co-development with customers. Relying on boundary theory and its wider contextualization in relation to resource dependence theory, the authors argue that an innovation-oriented strategy, combined with transformational and transactional leadership, represents senior management levers for driving general openness and providing strategic directions for innovation, which then helps bridge the boundary with customers and facilitate the co-development of new products. Data from 135 managers and 415 subordinates reveal that some senior management levers can foster co-development. Specifically, the results of the hierarchical regression analyses confirm the hypothesized positive interaction effect of innovation-oriented strategy and transformational leadership, whereas the combination of innovation-oriented strategy and transactional leadership is not beneficial. The strength of innovation-oriented strategy as an important senior management lever differs for goods and services, which is not the case for the other two levers. Finally, this study provides a more fine-grained perspective on the new product frequency outcomes of co-development, by relying on organizational learning theory. Most research is restricted to linear relationships, but the current investigation unveils an inverted U-shaped relationship between co-development and companies' new product frequency. Hence, companies can profit from co-development at lower and medium levels, but the common notion of "the more, the better" does not apply to an unlimited degree for co-development.

Practitioner Points

- Senior managers should implement an innovation-oriented strategy to provide guidance regarding the company's innovation generation goals in co-development projects.
- Companies need to support organizational members with adequate freedom and flexibility in terms of co-development projects, instead of focusing on fixed goals, expectations, or rewards.
- If they offer services in particular, companies should initiate appropriate training and coaching for senior managers, to help them better motivate and lead organizational members with a transformational leadership style.

- Co-development can be fostered by senior managers from the top but also should be tracked, using management tools and key performance indicators, to recognize negative efficiency trends as early as possible.

Introduction

Co-development with customers to generate new goods and services has become a strategic topic on virtually every company's agenda (Matthing, Sandén, and Edvardsson, 2004; Payne, Storbacka, and Frow, 2008; Prahalad and Ramaswamy, 2000), because it can contribute to better product-market fit and increase the chances of product success in the market (Gruner and Homburg, 2000; Neale and Corkindale, 1998). Furthermore, co-development—or the joint development of new products with customers through interaction and participation at various stages of the new product development process (Athaide, Stump, and Joshi, 2003; Carbonell, Rodríguez-Escudero, and Pujari, 2009; Hsieh and Chen, 2005)—can shorten new product development cycles and increase the speed to market, which eventually benefits

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downstream variables, such as company performance (Lau, Tang, and Yam, 2010; Sherman, Souder, and Jenssen, 2000). Forward-thinking companies thus integrate co-development into their business models. When Millennium Pharmaceuticals was founded, for example, senior management chose to focus solely on the early stages of drug development and foster partnerships with larger customer companies (e.g., Pfizer, Merck) to handle clinical trials and commercialization. As it grew, Millennium Pharmaceuticals' strategic evolution enabled it to leverage these co-development relationships in other parts of its operations (Chesbrough and Schwartz, 2007).

Despite the strategic relevance of co-development for practitioners, many companies remain uncertain about which levers can best foster its implementation. Extant research provides valuable insights into ways to organize successful co-development with customers (e.g., Athaide et al., 2003; Carbonell et al., 2009; Hoyer, Chandy, Dorotic, Krafft, and Singh, 2010), but suggestions for fostering co-development from the top are rare. This gap is surprising, considering the vast number of studies that highlight the important influence of senior managers on companies' innovativeness (e.g., Galasso and Simcoe, 2011; Yadav, Prahu, and Chandy, 2007) and performance (e.g., Bertrand and

Schoar, 2003; Iaquinto and Fredrickson, 1997). Although some studies note the influence of senior managers' leadership styles on innovation (e.g., Elenkov and Manev, 2005; Osborn and Marion, 2009), to the best of the authors' knowledge, no investigations address how senior managers might strategically support co-development. Yet strategy and leadership both drive innovation in general (Jung, Chow, and Wu, 2003; Yadav et al., 2007). By combining these concepts, the current study contributes to a deeper understanding of how companies can drive co-development from the top. Specifically, it addresses the following research question: *How does the interplay of strategy and leadership styles of senior managers relate to co-development?*

This study also includes the outcomes of co-development for new product frequency, which refers to the rate of introduction of new products (e.g., Katila, 2002; Stock and Zacharias, 2011). On the one hand, co-development can generate multiple benefits for companies, such as increased knowledge about customer needs (Fang, 2008; Narver, Slater, and MacLachlan, 2004), heightened efficiency (Hoyer et al., 2010; Payne et al., 2008), and improved innovation speed (Carbonell et al., 2009). Therefore, companies can introduce innovative products more frequently. On the other hand, companies confront increasing volumes of customer information when they engage in more co-development (Hoyer et al., 2010), which may decrease the speed of innovation and thus new product frequency. These arguments indicate that a simple logic, such as "the more co-development, the better or worse," is insufficient, and a nonlinear influence of co-development might exist. Thus, the second research question asks: *How is co-development related to new product frequency?*

Extant research emphasizes significant differences between goods and services, which may influence co-development processes too (e.g., Nijssen, Hillebrand, Vermeulen, and Kemp, 2006; Stock, 2011). Due to the unique characteristics of services—intangibility, inseparability, heterogeneity, and perishability (Lovelock and Gummesson, 2004)—their development process tends to be less formalized or standardized and thus requires more intensive information exchanges than the development process for goods (Ettlie and Rosenthal, 2011). The effectiveness of senior management levers may vary, depending on whether they are applied to the co-development of goods or services. Therefore, the third question this study seeks to answer is: *Are strategy and leadership styles differentially*

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associated with the co-development of goods and services?

In addressing these research questions, this study offers multiple contributions for research and managerial practice. First, it takes a top-down perspective on the senior management levers of co-development. Relying on boundary theory (Aldrich and Herker, 1977) and its wider contextualization in relation to resource dependence theory (Pfeffer and Salancik, 1978), this study investigates how senior management levers foster co-development. An innovation-oriented strategy, as “the degree to which a company’s strategy focuses on driving innovativeness” (Stock and Zacharias, 2011, p. 873), provides guidance for appropriate company behavior (Gatignon and Xuereb, 1997). In addition, senior managers might open their companies by providing transformational leadership, which involves intellectual stimulation, individualized consideration, and charisma (Bass, Avolio, Jung, and Berson, 2003; Howell and Avolio, 1993), or transactional leadership, which represents the extent to which senior managers clarify what they expect from employees (Jansen, Vera, and Crossan, 2009; Vaccaro, Jansen, Van Den Bosch, and Volberda, 2012). By investigating these senior management levers and their interactions, this article enriches the co-development field with knowledge about these largely neglected antecedents.

Second, this study provides a richer understanding of the new product frequency outcomes of co-development and sheds light on the underlying mechanism for the nonlinear relationship between co-development and the frequency of new product introductions. New product frequency is a particularly valuable measure in this context, because it augments the scientific discourse with a portfolio perspective on innovative products. In line with organizational learning theory (Levitt and March, 1988; March, 1991), new knowledge and the difficulty of disseminating this knowledge, by transforming it into new product development projects, results in a nonlinear impact of co-development.

Third, goods or services provide contingency factors that can alter the effects of senior management levers on co-development. The results reveal that the association of an innovation-oriented strategy with co-development is weaker in a service compared with a goods context. The related implications may be particularly useful for managerial practice, because across industries, companies constantly face pressures to develop new goods and services (Ali, Krapfel, and LaBahn, 1995; Atuahene-Gima, 1996).

Conceptual Background

Resource Dependence Theory

This study relies on resource dependence theory (Pfeffer and Salancik, 1978), according to which “organizations are not able to internally generate ... all the resources or functions required to maintain themselves” (Aldrich and Pfeffer, 1976, p. 83). Rather, companies are open systems that depend on their environments to acquire important resources for their value creation process, such as through customers (Pfeffer and Salancik, 1978). A common approach companies use to handle external dependence is to establish cooperative relationships with resource owners (i.e., customers), which helps reduce their dependence and increase their effectiveness and performance (Eggert, Ulaga, and Hollmann, 2009). Gruner and Homburg (2000) refer to such cooperative relationships as “bridging strategies.” Hillman, Withers, and Collins (2009) point out that the boundary conditions of resource dependence theory have been poorly examined. Co-development may represent a particularly important boundary condition for companies to bridge with customers, because with co-development, “the customer may also become active and cross the boundary into the provider sphere” (Grönroos and Voima, 2013, p. 141). Co-development helps ensure resource and information flows between partners, and it shifts tasks that were formerly conducted by suppliers to customers (Athaide and Klink, 2009; O’Hern and Rindfleisch, 2010).

Valuable insights (e.g., exchange norms, information processing, external representation for resource acquisition, adaption to influential environmental contingencies, and constraints) about how to bridge the boundary between suppliers and their customers come from boundary theory (Aldrich and Herker, 1977), which thus appears in prior research (e.g., Friedman and Podolny, 1992; Santos and Eisenhardt, 2005; Stock, 2006). Boundary theory suggests companies and organizational members should maintain interfaces with their suppliers, to obtain resource inputs from customers, such as information about their needs, ideas for new products, or support during the new product development process (Lau et al., 2010; Lysonski, 1985; Tushman, 1977). According to resource dependence theory in general and boundary theory in particular, proactively involving customers in new product development through co-development is an important activity for companies (Fang, 2008; Lau et al., 2010),

enabling them to interact with customers beyond the scope of basic market research (Carbonell et al., 2009; Hsieh and Chen, 2005). Boundary theory also identifies various factors that make it easier for companies to open themselves to co-development; strategy and leadership likely are important levers that support boundary roles (de Jong, de Ruyter, and Lemmink, 2004; Miller, 1987).

Strategy provides insights into the company's objectives, which can be broken down to a specific boundary condition (Aldrich and Herker, 1977). Leadership displayed by managers may provide openness in accordance with the company's overall goals, such as by guiding the organization to respond adequately to environmental requirements (Jemison, 1984; Osborn and Marion, 2009). In the current study context, strategy and leadership should create the organizational openness required to bridge the boundary to customers during co-development. These levers represent guidance from the top (Jemison, 1984; Osborn and Marion, 2009), supporting the company's openness to new perspectives and ideas from the customer (instead of exclusively internal insights; Emden, Calantone, and Droge, 2006). Finally, strategy and leadership help deal with the multiple interests of the involved parties and constructively solve any conflicts.

Study Framework

The proposed conceptual framework links three important senior management levers—innovation-oriented strategy, transformational leadership, and transactional leadership—and their interactions to co-development, which in turn are associated with new product frequency (see Figure 1). A company's long-term direction, with regard to its innovations, provides the general openness needed to bridge the boundary to customers through co-development. These long-term strategic goals constitute an *innovation-oriented strategy*. Companies following such a strategy focus on driving innovation generation and being first in new markets with their products (Miles, Snow, Meyer, and Coleman, 1978; Stock and Zacharias, 2011). This innovation-oriented strategy supports the generation of innovations through active searches for new opportunities and continuous creations of new products (Gatignon and Xuereb, 1997). Hence, highly innovation-oriented companies may be more likely to co-develop with their customers to gain insights into their needs and ideas and generate new products frequently.

By providing guidance from the top through leadership, companies also can open themselves to new insights, environmental requirements, perspectives, overall goals, and ideas from the customer, instead of exclusively considering internal insights (Emden et al., 2006; Jemison, 1984; Osborn and Marion, 2009). Senior managers' *transformational leadership* style may provide openness by expressing their vision, allowing individual development, and leveraging subordinates' motivation to develop promising new innovative products creatively with customers, in an environment of multiple (sometimes contradictory) interests (Bass et al., 2003; Boerner, Eisenbeiss, and Griesser, 2007; Fang 2008; Howell and Avolio, 1993; Pieterse, Van Knippenberg, Schippers, and Stam, 2009; Stock, 2006). Thus, senior managers' transformational leadership style can stimulate subordinates' openness to idea generation, performance beyond expectations, independent thinking, and willingness to take on more responsibility (Bass, 1990; Boerner et al., 2007), which should support co-development.

In addition, openness can stem from senior managers' *transactional leadership* (Jansen et al., 2009; Vaccaro et al., 2012), in which they promote an open, feedback-based relationship by clearly expressing expectations, setting goals, and clarifying the link between performance and rewards (Bass, 1990; Bass et al., 2003; Jansen et al., 2009). Senior managers providing transactional leadership constantly monitor organizational members' actions, to anticipate mistakes before they arise and take corrective actions when required (Howell and Avolio, 1993). They also do not hesitate to punish organizational members for deviating from those standards or missing performance targets (Bass et al., 2003). Senior managers providing transactional leadership thus convey a clear understanding of what is expected of organizational members during co-development with customers, such that transactional leadership may represent a lever to emphasize companies' openness and provide direction in co-development projects.

In the next stage of the framework, co-development connects with *new product frequency*, defined as the rate of introduction of new products (e.g., Katila, 2002; Stock and Zacharias, 2011). The choice of this dependent variable reflects studies that focus on the frequency of new products as an outcome measure for innovation generation (e.g., Dibrell, Craig, and Hansen, 2011; Stock, Six, and Zacharias, 2013). Companies that frequently introduce new products often can

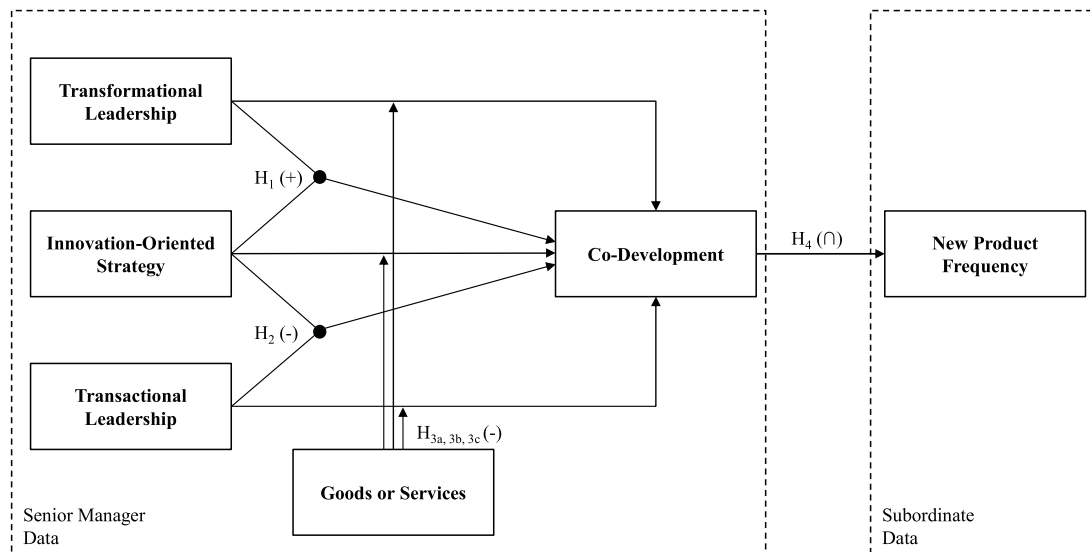


Figure 1. Conceptual Framework

offset losses due to failures, because their revenue flow is high enough.¹

In briefly describing the three senior management levers and how they relate to co-development, the linkages among the three levers are relatively plausible, such that they do not require explicit hypotheses. Rather, the focus in the hypotheses development is on the interactions of strategy and leadership styles, as well as on how the strength of the relationship of the three senior management levers with co-development might vary for goods versus services. This section concludes with a detailed theoretical foundation for the relationship between co-development and new product frequency.

Interaction Effects: Innovation-Oriented Strategy and Leadership Styles

Implementing an innovation-oriented strategy can help companies excite their customers, outperform competitors, enter markets first, and build an innovative product portfolio (Bowonder, Dambal, Kumar, and Shirodkar, 2010; Stock and Zacharias, 2011). From a boundary theory perspective, an innovation-oriented strategy can provide strategic directions for innovation, to bridge the boundary with customers and facilitate the co-development of new products. Senior managers' transformational leadership style may foster the implementation of such an innovation-oriented strategy for

the benefit of co-development, because it increases the organization's openness by providing a vision, inspiration, and autonomy, which support the creation of systems and structures at all levels of the organization (Bryant, 2003).

Senior managers with a transformational leadership style concentrate on developing a vision and inspiring followers to pursue the company's long-term goals (Jung et al., 2003). Transformational leaders focus on actively promoting organizational members' participation in collective decisions and activities (Aragón-Correa, García-Morales, and Cordon-Pozo, 2007; Bass, 1990), thereby motivating the members to get involved and defer their own self-interests so that they can achieve the organization's mission and gain approval or esteem from senior managers (Bass, 1990; Bryant, 2003; Elenkov and Manev, 2005).

Senior managers who provide transformational leadership also motivate organizational members to accept the senior managers' values as their own, such as being innovative and creative or sharing their ideas with the company (Bryant, 2003; Elenkov and Manev, 2005). This encouragement supports the successful implementation of an innovation-oriented strategy. An innovation-oriented strategy thus should be a more effective driver of co-development for managers who lead in a transformational way. Organizational members' strong awareness of the company's long-term goals, missions, and visions, due to transformational leadership, shifts their focus from short-term, immediate solutions or objectives to a long-term perspective (Jung et al., 2003). Then it is more natural for these

¹Extant literature also indicates that greater development frequency pays off in the form of shorter cycle times (Ali et al., 1995), market share/value (Chaney and Devinney, 1992), sales growth, profitability (Sandvik and Sandvik, 2003), and company survival (Han, Kim, and Srivastava, 1998).

members to follow an innovation-oriented strategy that emphasizes co-development. Overall, senior managers' transformational leadership provides vision, inspiration, and autonomy to organizational members, which seems appropriate for implementing an innovation-oriented strategy in co-development projects.

H1: The interaction between an innovation-oriented strategy and transformational leadership is positively associated with co-development.

Senior managers with a transactional leadership style instead provide relatively concrete directions to organizational members, monitor exchange relationships, and maintain the status quo rather than leading change (de Hoogh, den Hartog, and Koopman, 2005; Jansen et al., 2009). Thereby, these leaders “exercise a maintenance role, ... reinforce existing strategies, focus on increasing efficiency in current practices, and communicate the benefits of incremental refinements to existing innovation trajectories” (Jansen et al., 2009, p. 9). Such senior managers do not provide a flexible framework for the development and implementation of an innovation-oriented strategy, which would require a new way of thinking and willingness to change.

Considering modern managers' daily routines, it is questionable whether senior managers with a transactional leadership style can create an atmosphere that enables organizational members to follow an innovation-oriented strategy and foster co-development with customers. Rather, overly concrete directions from transactional leaders may make it more difficult for organizational members to readjust their activities in co-development projects, according to the changing circumstances and requirements of those projects. Existing directions by senior managers might conflict with new situations, which may unsettle organizational members, reduce their openness, and limit their willingness to follow or implement an innovation-oriented strategy through co-development (Fang, 2008; Fliess and Becker, 2006).

Finally, transactional leaders are responsive to organizational members' self-interests only to the extent that they can be met while still getting the work done. As a consequence, organizational members are not motivated to give anything beyond what is clearly specified (Bryant, 2003), which cannot effectively support an innovation-oriented strategy. Therefore, an innovation-oriented strategy seems less effective for driving co-development in an organization that features a transactional leadership style. The “give-and-take” approach adopted by transactional

leaders may not be appropriate to implement and follow an innovation-oriented strategy in the long run, which requires organizational members to embrace the innovation concept.

H2: The interaction between an innovation-oriented strategy and transactional leadership is negatively associated with co-development.

Contingency Effects of Goods Versus Services

Regarding potential moderating effects on the links between senior management levers and co-development, it is pertinent to distinguish between co-developed goods or services (Nijssen et al., 2006; Stock, 2011). Services differ from goods in their intangibility, inseparability, heterogeneity, and perishability (Lovelock and Gummesson, 2004). Therefore, the development process for new services is usually less formalized or standardized than the development process for goods (Ettlie and Rosenthal, 2011), and firms may focus more on improvisation and emergent strategies (Mintzberg and Waters, 1985; Moorman and Miner, 1998). In turn, the relationship between senior management levers and co-development may be weaker for services than for goods.

The intangibility of services also requires more intensive information exchanges between the supplier company and its customers, so organizational members must focus relatively more on customer needs than on their senior managers' leadership (Gummesson, 2002; Lievens and Moenaert, 2000). Services also are mainly developed through ad hoc decisions, made during customer interactions, so companies need to react more rapidly in a service context than in a goods context to secure a competitive advantage (Atuahene-Gima, 1996). The integration of senior managers' requirements might slow down and hinder this interaction (Atuahene-Gima, 1996; Droege, Hildebrand, and Heras Forcada, 2009). Finally, services are often developed quickly, get introduced into the market for testing, and then are developed further in the market (de Brentani, 1995), so customer and market demands likely dominate further service development, rather than senior managers' guidance.

Due to these characteristics of services, especially as they affect new product development processes, service companies and their organizational members may be less receptive to guidance from the top, compared with their counterparts in goods sectors (Lievens and Moenaert, 2000). Therefore, senior management levers may be relatively less effective in fostering co-

development for new services (Lievens and Moenaert, 2000).

H3: The relationships of (a) innovation-oriented strategy, (b) transformational leadership, and (c) transactional leadership with co-development are not as strong for services as they are for goods.

New Product Frequency Implications of Co-development

Finally, to explain the relationship between co-development and new product frequency, organizational learning theory (Levitt and March, 1988; March, 1991) relates to the development of new knowledge or insights that influence organizational behavior (Huber, 1991). Co-development aims to bring new insights into the organization, for incorporation into the new product development processes. From an organizational learning perspective, companies gain *new knowledge* by developing it jointly with their customers (Slater and Narver, 1995). The creation of new knowledge through organizational learning thus is critical for innovation activities (García-Morales, Jiménez-Barrionuevo, and Gutiérrez-Gutiérrez, 2012), particularly knowledge from outside the company (Jiménez-Jiménez and Sanz-Valle, 2011). Although new knowledge might be put immediately to use by organizational members, it also must be disseminated and shared throughout the organization to realize its full potential (Cohen and Levinthal, 1990). Thus, co-development supports new knowledge gains but also necessitates greater integration efforts to ensure the dissemination and sharing of this new knowledge.

Co-development offers great potential to acquire knowledge from outside the company, related to new product ideas and information about customer needs (Hsieh and Chen, 2005; O'Hern and Rindfleisch, 2010). When companies co-develop more intensively with customers, they gain customer-driven inputs for co-development, which helps them quickly identify customer needs and bring new products to market. Therefore, companies' innovation activities should speed up due to co-development, which may lead to higher new product frequency.

Beyond a certain level though, increasing co-development further could impede new product frequency. That is, with increasing levels of co-development, it becomes more difficult for companies

to apply the new knowledge they are obtaining to commercial ends (Cohen and Levinthal, 1990; Jiménez-Jiménez and Sanz-Valle, 2011). At lower levels of co-development, expressed customer needs find their way into the new product development process fairly easily (Hoyer et al., 2010). When co-development increases from medium to high levels, the still unaddressed customer needs may be more difficult to commercialize though, because they are more complex and specific and thus more difficult to interpret (Bettencourt, Ostrom, Brown, and Roundtree, 2002). At higher levels of co-development, companies also must deal with much more information, making it more complicated to distill the useful, relevant information (Hoyer et al., 2010).

In this case, organizational members have to deal with an information overload, which can reduce their performance, decision quality, and effectiveness (Chan, 2001; O'Reilly, 1980; Snowball, 1980). Considering the amount of information involved in many co-development projects, companies actively seek to manage the flow and analyses of this information. Other important tasks, such as tracking actual market developments, may recede to the background. If companies need to focus on fewer projects with more complex information, it eventually might reduce their new product frequency. Organizational learning theory also indicates that more complex information makes interpretation more difficult, raising the effort and resources required to make use of that information (Huber, 1991). As co-development increases from low to moderate levels, the advantage of acquiring new knowledge grows, and so does new product frequency. Beyond a certain point though, increasing co-development lowers the benefits of new knowledge, due to its complexity, and new product frequency declines.

H4: Co-development exhibits an inverted U-shaped relationship with new product frequency.

Methodology

Data Collection and Sample

This study is based on multi-informant data, obtained from an international senior manager MBA program. First, emails were sent to 298 senior managers attending the MBA program, asking them to participate in a survey study regarding the influence of strategy and

Table 1. Sample Composition

Company Data			
Industry Sector		Number of Employees	
Manufacturing	23.7%	<100	18.5%
Professional Services	22.2%	101–500	22.2%
IT	15.6%	501–1,000	10.4%
Banking	13.3%	1,001–2,500	14.1%
Retail	12.6%	2,501–5,000	7.4%
Machinery & electronics	7.4%	5,001–10,000	4.4%
Equipment suppliers	5.2%	10,001–20,000	5.2%
Sales (Annual Volume)		20,001–50,000	5.9%
		50,001–100,000	8.2%
<\$5 million	7.4%	>100,000	3.7%
\$6–\$10 million	6.7%		
\$11–\$15 million	4.4%		
\$16–\$25 million	12.6%		
\$26–\$50 million	6.7%		
\$51–\$100 million	12.6%		
\$101–\$200 million	6.7%		
\$201–\$500 million	6.7%		
\$501–\$1 billion	13.3%		
>\$1 billion	22.9%		

leadership on the innovation generation of their companies. Second, each of the senior managers received a written questionnaire to complete. Third, to support the multi-informant design, participants provided email addresses for at least five subordinates working under their direct authority or with a direct reporting function to them.

The questionnaires to subordinates were sent to them separately, with a postage-paid return envelope. The subordinates were told that their individual data would remain confidential and that only aggregated results would be used for this study. The senior manager data were matched with the subordinate data according to a common identification code on the questionnaire. Nonresponse and missing data reduced the final sample to 135 senior managers (response rate = 45.3%) and 415 subordinates (response rate = 33.1%), for an average of 3.1 subordinate assessments per senior manager. The distribution of subordinates per senior manager is balanced, with a standard deviation of only .62. The participating senior managers held positions as CEOs, CIOs, CFOs, board members, upper management, and others. All of them had profit-and-loss responsibility. Their mean age was 38.5 years, and they worked in a wide range of industries: manufacturing (23.7%), professional services (22.2%), IT (15.3%), banking (13.3%), retail (12.6%), machinery and electronics (7.4%), and equipment suppliers (5.2%).

The co-developed goods and services in this sample include large machines, such as printers, blanking, and pressing machines; software and communication

network solutions; new flavors and colors for fruit juices; controllers for robots; concepts for property valuation; and customized product supply systems. The number of employees in the companies ranged from less than 100 to more than 100,000, and the sales volumes ranged from less than \$5 million to more than \$1 billion per year. Table 1 provides supplemental information about the sample.

Measure Development and Properties

Scale development. Standard scale development procedures were applied, including a comprehensive literature review. When possible, existing scales were modified or further developed to match the study context. The resulting questionnaire was pretested and refined; all items (including sources for scale development) are detailed in the Appendix.

Measures. The measure of innovation-oriented strategy featured a reflective, three-item scale adopted from Stock and Zacharias (2011). Both transformational and transactional leadership were operationalized as formative second-order constructs, each comprising several first-order reflective variables, as recommended by MacKenzie, Podsakoff, and Jarvis (2005) and applied in extant research (e.g., Menguc, Auh, and Shih, 2007; Pieterse et al., 2009). The first-order constructs constituting the transformational leadership variable were operationalized with scales adapted from the multifactor leadership questionnaire (MLQ) by Bass and Avolio (1995). Intellectual stimulation and individualized consideration of a senior manager were measured by four reflective items each, and charisma was measured by three reflective items. With regard to transactional leadership, the first-order constructs were contingent rewards and active management by exception (Hetland, Sandal, and Johnsen, 2008; Pieterse et al., 2009; Waldman, Ramirez, House, and Puranam, 2001); a third potential dimension, passive management by exception (Howell and Avolio, 1993), was excluded on the basis of qualitative interviews with CEOs and senior managers of supplier firms in advance of the survey. These interview partners shared their experiences in co-development projects, which revealed that passive management by exception, or the timely intervention of a leader after mistakes and failures (Howell and Avolio, 1993), does not fit a co-development context. Instead, the focus for this measure is on contingent rewards and active management by exception, operationalized by four reflective items

each from the MLQ (Bass and Avolio, 1995). Exploratory and confirmatory factor analyses eliminated items that did not meet or caused constructs not to meet established criteria for reliability and validity, including one item from the intellectual stimulation and charisma measures (transformational leadership) and one item from the measures of contingent rewards and active management by exception (transactional leadership). The final scales appear in the Appendix.

The co-development construct was adapted from Hsieh and Chen (2005) and measured by five reflective items, adjusted to fit the research context. The operationalization of new product frequency relied on five reflective items from Stock and Zacharias (2011).

The senior management levers—innovation-oriented strategy, transformational, and transactional leadership—and co-development all were assessed by senior managers, who have a good overview of the whole company and are knowledgeable about its actual and future strategy. However, because senior managers often exhibit overconfidence, narcissism, or hubris, they tend to overestimate their own performance and success indicators for which they are responsible (e.g., Galasso and Simcoe, 2011; Gerstner, König, Enders, and Hambrick, 2013; Simon and Houghton, 2003; Tang, Li, and Yang, 2012). Therefore, the senior managers' subordinates evaluated new product frequency as an outcome of co-development; they should be more neutral and less biased.

The moderating effect of goods or services used a dummy-coded variable based on industry sector information and the main offerings of the participating companies, validated with a review of the specific goods and services offered by each company. On the basis of this information, the companies could be assigned to categories: manufacturing, IT (which mainly included manufacturing aspects, such as solutions for communication networks, server building, and hardware components), machinery and electronics, and equipment suppliers constituted the goods category, and professional services, banking, and retail were the service category.

Various control variables, assessed by senior managers, helped ensure the validity of the results. These variables described company characteristics (number of employees, sales, and capital intensity) and the senior managers' characteristics (organizational tenure in years, age, and an effect-coded variable to represent their position in the company). Several environmental factors also might be relevant: competitive intensity operationalized by five reflective items and

technological turbulence operationalized by four reflective items (Jaworski and Kohli, 1993). As an additional control variable for the nonlinear relationship of co-development with new product frequency, traditional customer information acquisition was measured by five reflective items adapted from Jaworski and Kohli (1993) and Li and Calantone (1998). This variable represents companies' "traditional" market research activities, used to identify customer needs (e.g., focus groups, phone interviews, surveys; Matthing et al., 2004). Prior research indicates both positive (Ordanini and Parasuraman, 2011; Parry and Song, 2010) and negative (Hoyer et al., 2010; Matthing et al., 2004) relationships of traditional customer information acquisition with new product frequency; their relation may be nonlinear.

Measurement properties. Exploratory and confirmatory factor analyses checked the reliability and validity of the scales (e.g., Gerbing and Anderson, 1988). The Cronbach's alphas exceeded the recommended minimum of .7 for all reflective constructs (Nunnally, 1978). Composite reliability was greater than the threshold value of .6 for all constructs (Bagozzi and Yi, 1988). In terms of the average variance extracted (AVE), all constructs in the sample exceeded the minimum requirement of .5 (Bagozzi and Yi, 1988). The test for the discriminant validity of the reflective constructs measured with multiple items used Fornell and Larcker's (1981) criterion. As Table 2 reveals, the diagonal elements representing the square roots of the AVE were greater than the off-diagonal bivariate correlations with other constructs, which confirmed the conceptual distinction of these factors.

A check of the subordinate data tested whether aggregating the assessments by groups of subordinates for each senior manager was appropriate, according to the index of within-group interrater reliability (r_{wg}) established by James, Demaree, and Wolf (1984). The median r_{wg} value for new product frequency was .93, exceeding the minimum of .7 (Burke, Finkelstein, and Dusig, 1999) and justifying the data aggregation. Thus the averaged subordinate responses for each senior manager formed a single-group composite value (van Bruggen, Lilien, and Kacker, 2002).

Evaluation of Potential Biases

Although multicollinearity may threaten the validity of the results (Cortina, 1993), the largest variance inflation

Table 2. Correlations and Descriptive Statistics

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. Innovation-oriented strategy	.748																
2. Transformational leadership	.360**	N/A															
3. Transactional leadership	.334**	.437**	N/A														
4. Co-development	.465**	.348**	.274**	.787													
5. New product frequency	.289**	.173*	.272**	.235*	.866												
6. Number of employees	-.024	.030	-.109	.104	.103	N/A											
7. Sales	-.043	.055	-.127	.044	.146	.648**	N/A										
8. Technology turbulence	.081	.081	.023	.052	.035	.103	-.013	.787									
9. Competitive intensity	-.110	.027	-.105	-.062	-.146	.234**	.257**	.059	.721								
10. Capital intensity	-.043	.054	.029	-.085	-.026	.061	.245**	-.037	.095	N/A							
11. Organizational tenure	.087	.101	.070	.073	.056	.085	.101	-.113	-.031	-.183*	N/A						
12. Age of senior manager	.098	.077	.011	.154	.005	-.037	.072	-.111	-.159	.111	.340**	N/A					
13. SM position 1	-.001	.122	.180*	-.071	.112	-.268**	-.166	-.122	-.072	-.027	.092	.082	N/A				
14. SM position 2	-.145	-.031	-.083	-.043	.044	.257**	.328**	.028	-.019	.027	-.005	-.095	.266**	N/A			
15. SM position 3	-.090	.052	-.033	.054	.128	.049	.170*	-.055	.053	.048	-.004	-.023	.628**	.658**	N/A		
16. TCIA	.491**	.331**	.276**	.525**	.154	.297**	.251**	.077	.055	-.025	-.063	.002	-.172*	-.012	-.063	N/A	
17. Goods or services	.056	.017	.047	.074	.030	-.050	.066	-.248**	-.001	.070	-.145	-.069	-.024	-.045	-.029	.086	N/A
Mean	4.733	5.500	5.224	4.764	5.200	4.080	6.480	4.130	4.273	4.150	8.172	38.50	.044	.541	-.096	5.053	.482
Standard deviation	1.318	.598	.736	1.269	.881	2.783	2.880	1.436	1.379	1.991	5.091	5.522	.544	.710	.384	1.265	.502

* $p < .05$. ** $p < .01$.

Notes: $N = 135$. The bold diagonal elements are the square roots of the average variances extracted for reflective constructs measured with multiple items; SM = senior manager; SM position 1 = CEO; SM position 2 = CIO, CFO, board member, upper management; SM position 3 = others; TCIA = traditional customer information acquisition.

factor reached a value of 3.3, well below the cutoff of 10 (Mason and Perreault, 1991). Multicollinearity thus did not appear to be a problem. As a first test for potential common method bias, Harman's single-factor test was applied (Podsakoff, MacKenzie, Lee, and Podsakoff, 2003). The results indicated eight separate factors, and no single factor accounted for more than 25.1% of the covariance among the modeled variables, similar to comparable prior literature (e.g., Aragón-Correa et al., 2007; Scott and Bruce, 1994). That is, no single factor emerged, nor did one factor account for most of the variance. Following Lindell and Whitney (2001), a marker variable test also compared the observed correlations between constructs with correlations adjusted for the effect of sales as a marker variable, which had the lowest correlation with the key co-development variable ($r = .044$; Lindell and Brandt, 2000). The correlations of the key variables remained significant after the correction. Both tests, together with the use of multi-informant data, suggested common method bias was unlikely to be a problem.

With regard to potential endogeneity, prior literature offers some possible sources, such as missing variables, false measurement, autocorrelation, simultaneity, and self-selection problems (Bascle, 2008; Cornwell and Trumbull, 1994; Hamilton and Nickerson, 2003). In the proposed research model, missing variables and self-selection are potential concerns, though the relevant control variables in the model minimize the potential risk of endogeneity due to missing variables. With regard to self-selection, prior literature recommends using panel data (e.g., Hamilton and Nickerson, 2003; Polachek, 1981; Semykina and Wooldridge, 2010), which were not available for this study. Instead, to avoid potential group membership through self-selection and ensure data heterogeneity, the study participants were selected systematically. Overall then, endogeneity does not appear to be a major concern for this study.

Hypotheses Tests

The test of the quadratic relation of co-development with new product frequency requires an appropriate method. Hierarchical regression analysis is a widely used method to estimate nonlinear relationships among variables, because it can examine an overall main effect by simultaneously including the linear and quadratic terms of variables in analyses (e.g., Aiken and West, 1991; Kohtamäki, Partanen, Parida, and Wincent, 2013;

Stock, 2011). Therefore, the hypotheses tests relied on two hierarchical regression analyses with standardized constituent variables (Aiken and West, 1991; Tabachnick and Fidell, 2013). Regression analysis A tested for the main effects of senior management levers on co-development, the interaction effects between strategy and leadership styles, and the moderating effects of goods or services. Regression analysis B instead provides results for the nonlinear relationship of co-development with new product frequency.

The first step of regression analysis A included only control variables (Model 1). Then the second step (Model 2) added the independent senior management levers (innovation-oriented strategy, transformational leadership, and transactional leadership). Model 3 included the interaction effects of innovation-oriented strategy and transformational/transactional leadership. Adding the interaction effects of the goods or services moderator to Model 2 formed Model 4. To prepare for regression analysis B, the linear term of co-development was squared to assess a potential nonlinear relationship with new product frequency (Aiken and West, 1991). Regression analysis B then started with the control variables entered in a first step (Model 1), before adding the linear term of co-development in a second step (Model 2) and the squared term of co-development in a third step (Model 3).

Results

Regression analysis A provides results related to the main, interaction, and moderating effects of the framework. As Table 3 shows, the adjusted *R*-square values of Models 3, 4, and 5 increase significantly compared with Model 2, resulting in 35.5% in Model 3 (interaction model), 33.5% in Model 4 (moderator model), and 40.2% in Model 5 (full model) for the co-development dependent variable.

Innovation-oriented strategy and leadership styles exert important influences on co-development with customers. Model 2 of regression analysis A provides the empirical results for the nonhypothesized main effects. An innovation-oriented strategy is positively associated with co-development ($\beta = .343$; $p < .01$). Senior managers can clarify long-term understanding and beliefs about innovation within the organization by providing strategic direction. With regard to the relationship between transformational leadership and co-development, the results again reveal a positive association ($\beta = .171$; $p < .05$). That is, senior

Table 3. Results of Hypotheses Tests

Dependent Variable	Regression Analysis A Co-Development					Regression Analysis B New Product Frequency		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 1	Model 2	Model 3
<i>Control Variables</i>								
Number of Employees	.137	.126	.133	.088	.083	-.012	-.015	-.019
Sales	-.036	-.066	-.055	-.027	-.009	.205	.231	.250*
Technology turbulence	.058	.010	.069	.086	.130	.042	.036	.029
Competitive intensity	-.091	-.037	-.088	-.044	-.090	-.207*	-.194*	-.192*
Capital intensity	-.106	-.121	-.036	-.117	-.031	-.023	-.013	-.004
Organizational tenure	.014	-.024	.036	-.045	.017	.078	.061	.082
Age of senior manager	.166	.159	.106	.147	.109	-.120	-.147	-.187*
SM position 1	-.224	-.316**	-.276**	-.271*	-.230*	.117	.155	.119
SM position 2	-.230	-.139	-.136	-.128	-.131	-.154	-.122	-.128
SM position 3	.363*	.390**	.388**	.349**	.346**	.153	.084	.142
TCIA						.259*	.146	.189
TCIA squared						.244*	.234*	.351**
<i>Main Effects Analysis A</i>								
Goods or services		.071	.102	.087	.113			
Innovation-oriented strategy		.343**	.335**	.320**	.312**			
Transformational leadership		.171*	.116	.169*	.107			
Transactional leadership		.144	.132	.151	.144			
<i>Interaction effects analysis A</i>								
Innovation-oriented strategy × transformational leadership			.317**		.291**			
Innovation-oriented strategy × transactional leadership			-.250**		-.279**			
<i>Moderating effects analysis A</i>								
Innovation-oriented strategy × goods or services				-.277**	-.255**			
Transformational leadership × goods or services				.044	-.032			
Transactional leadership × goods or services				.011	.054			
<i>Main Effects Analysis B</i>								
Co-development							.197	.126
Co-development squared								-.256**
R^2	.102	.359	.432	.419	.487	.161	.186	.233
Adjusted R^2	.030	.284	.355	.335	.402	.078	.098	.143
F -Value	1.415	4.804**	5.616**	4.972**	5.750**	1.948*	2.122*	2.602**
Incremental R^2	.102	.257	.073 ^a	.060 ^a	.128 ^a	.161	.025	.047
F -value for incremental R^2	1.415	12.019**	7.601** ^a	4.047** ^a	5.742** ^a	1.948*	3.698*	7.381**

* $p < .05$. ** $p < .01$.

Notes: $N = 135$. Standardized regression coefficients are reported. Two-tailed tests. SM = senior manager; SM position 1 = CEO; SM position 2 = CIO, CFO, board member, upper management; SM position 3 = others; TCIA = traditional customer information acquisition.

^aValues of incremental R^2 and F -value for incremental R^2 of Model 3, Model 4, and Model 5 refer to Model 2 of regression analysis A.

managers' transformational leadership style is effective for supporting organizational members' openness in terms of bridging the boundary to the customer during co-development, by expressing a vision, allowing individual development, and supporting an open culture. For the relationship between transactional leadership and co-development, the results do not indicate a significant association.

The analysis of the interaction effects in Model 3 confirms H1 and H2. In support of H1, which suggests that

the interaction effect of innovation-oriented strategy and transformational leadership is positive, the interaction is positive ($\beta = .317$; $p < .01$). Thus, a transformational leadership style together with an innovation-oriented strategy seems appropriate for co-development. The support for H2, which suggests a negative interaction of innovation-oriented strategy and transactional leadership ($\beta = -.250$; $p < .01$), indicates that a transactional leadership style combined with an innovation-oriented strategy instead is detrimental to co-development.

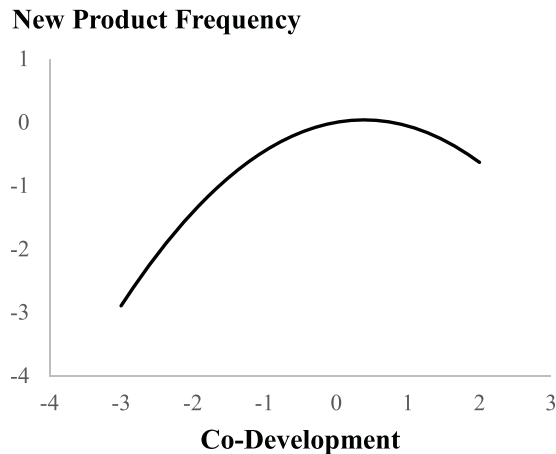


Figure 2. Relationship between Co-Development and New Product Frequency

Regarding the moderating effects, the results partially confirm H3a, H3b, H3c. H3a suggests that the relationship between an innovation-oriented strategy and co-development is not as strong for services as it is for goods, and a negative moderating relationship confirms the prediction ($\beta = -.277$; $p < .01$). Due to the characteristics of services, the implementation of an innovation-oriented strategy is less supportive of co-development activities in a service context than in a goods context. H3b and H3c suggest weaker relationships of transformational and transactional leadership with co-development for services than for goods, but no empirical support emerges for these predictions.

Regression analysis B provides the results regarding the nonlinear relationship of co-development with new product frequency. Moving stepwise from Models 1 to 3, Table 3 shows that the adjusted R -square values increase significantly with each step, resulting in 9.8% in Model 2 and 14.3% in Model 3. Model 2 indicates no significant association of the linear relationship of co-development with new product frequency. The nonlinear relationship of co-development in Model 3 instead reveals a negative, significant association ($\beta = -.256$; $p < .01$). The relationship between co-development and new product frequency thus takes an inverted U-shape, as predicted in H4.

These empirical results are plotted in Figure 2, which reflects an equation with standardized regression coefficients without an intercept. The x -axis depicts the range from the mean of co-development minus 3 standard deviations to plus 2 standard deviations, which nearly covers the seven-point Likert scale of co-development in the survey. In line with Atuahene-Gima, Slater, and Olson (2005), just the shape of the

curve and its slope at specific points are meaningful. The partial derivative indicates a maximum at $\partial f(\text{co-dev})/\partial \text{co-dev} = .126 - .512 \times \text{co-dev} = 0$, which occurs when $\text{co-dev} = .246$ on a standardized scale—that is, about one-quarter of a standard deviation above the mean of co-development. At the average value of co-development, the function thus is still increasing, and indicating that more co-development at this particular point will still increase new product frequency (before it starts to drop at about one-quarter of standard deviation above the mean).

To conclude the analyses, the bootstrapping approach suggested by Hayes and Preacher (2010) tests for mediation in the research model, by quantifying and examining the indirect effects of the senior management levers on new product frequency. This approach supports an estimation of the indirect effects in models of mediators and outcomes that are nonlinear functions, as is the case for the proposed model. Because an indirect effect cannot be quantified as a constant value if one of the constituent paths is nonlinear, it must be assessed for specific, representative values of the independent variable (Hayes and Preacher, 2010). Therefore, three separate bootstrap analyses assess the indirect effects of the three independent variables on new product frequency, using representative values (in .5 steps, below and above their corresponding means, within the scale range) for innovation-oriented strategy, transformational leadership, and transactional leadership. Each bootstrap analysis was executed with 1000 resamples, as recommended by Preacher and Hayes (2008), and the indirect effects were estimated within 95% confidence intervals (Zhao, Lynch, and Chen, 2010).

The results in Table 4 offer support for an indirect effect of innovation-oriented strategy and transformational leadership on new product frequency through co-development. The representative value estimates change, from a positive sign for low values of the independent variable to a negative sign for higher values. Thus, the relevant confidence intervals at the lower and upper ends of the examined values differ significantly from 0 at a .05 significance level. For transactional leadership, the findings offer no support for an indirect effect on new product frequency, because the relevant confidence intervals did not differ significantly from 0.

Several supplemental analyses provide robustness checks. For example, the inverted U-shaped relationship between co-development and new product frequency holds for all industries in the study sample. No industry had a significant moderating influence on the U-shaped

Table 4. Results of Mediation Tests

Indirect effects	Lower 2.5%	Estimate	Upper 2.5%
<i>Innovation-Oriented Strategy on New Product Frequency through Co-Development</i>			
Innovation-oriented strategy = −3.5	.011	.260	.649
Innovation-oriented strategy = −3.0	.001	.222	.554
Innovation-oriented strategy = −2.5	−.004	.184	.465
Innovation-oriented strategy = −2.0	−.005	.146	.391
Innovation-oriented strategy = −1.5	.004	.108	.355
Innovation-oriented strategy = −1.0	−.009	.070	.248
Innovation-oriented strategy = −.5	−.019	.032	.181
Innovation-oriented strategy = 0	−.064	−.006	.113
Innovation-oriented strategy = .5	−.123	−.044	.045
Innovation-oriented strategy = 1.0	−.214	−.082	−.014
Innovation-oriented strategy = 1.5	−.315	−.120	−.028
Innovation-oriented strategy = 2.0	−.389	−.185	−.038
<i>Transformational Leadership on New Product Frequency through Co-Development</i>			
Transformational leadership = −4.0	.013	.194	.724
Transformational leadership = −3.5	.010	.170	.631
Transformational leadership = −3.0	.006	.146	.539
Transformational leadership = −2.5	−.001	.122	.446
Transformational leadership = −2.0	−.002	.098	.353
Transformational leadership = −1.5	−.012	.074	.261
Transformational leadership = −1.0	−.014	.050	.168
Transformational leadership = −.5	−.052	.026	.085
Transformational leadership = 0	−.041	.002	.050
Transformational leadership = .5	−.086	−.022	.028
Transformational leadership = 1.0	−.132	−.046	.007
Transformational leadership = 1.5	−.175	−.070	−.010
<i>Transactional Leadership on New Product Frequency through Co-Development</i>			
Transactional leadership = −4.0	−.020	.160	.418
Transactional leadership = −3.5	−.012	.140	.415
Transactional leadership = −3.0	−.013	.119	.347
Transactional leadership = −2.5	−.005	.098	.278
Transactional leadership = −2.0	−.016	.078	.228
Transactional leadership = −1.5	−.023	.057	.192
Transactional leadership = −1.0	−.032	.036	.195
Transactional leadership = −.5	−.037	.016	.125
Transactional leadership = 0	−.058	−.005	.088
Transactional leadership = .5	−.095	−.026	.059
Transactional leadership = 1.0	−.135	−.046	.031
Transactional leadership = 1.5	−.221	−.067	.019

Notes: Bias-corrected confidence intervals for indirect effects are reported.

relationship, and the main results stayed robust. To exclude a possible selection bias, a test that included the number of subordinates responding as an additional control variable in regression analysis B showed that this variable had no significant relationship ($\beta = -.002$; *n.s.*), so a selection bias was unlikely.

Discussion and Conclusions

In line with resource dependence and boundary theory, an innovation-oriented strategy and a transformational leadership style foster the organizational openness and awareness necessary for co-development, better identify demands and dependencies in the environment, and build and maintain interfaces with customers

(Lyonski, 1985; Tushman, 1977). The gap between the company and customers thus can be bridged, which supports better coordination and the acquisition of needed resources, as well as increased motivation among organizational members to question assumptions and reframe problems in new ways (Bass et al., 2003; Howell and Avolio, 1993; Manev and Stevenson, 2001). Because co-developing parties usually interact intensively and frequently, support for organizational members' self-confidence through transformational leadership seems more effective than clearly expressed expectations, goals, and links between performance and rewards, as provided by transactional leadership (Bass et al., 2003; Jansen et al., 2009). This finding resonates with literature that investigates the

relationship of transactional leadership with innovation generation (e.g., Oke, Munshi, and Walumbwa, 2009; Pieterse et al., 2009).

That is, when senior managers follow a transactional leadership style, they provide relatively concrete directions to organizational members and tend to reinforce existing strategies by maintaining the status quo rather than leading change (de Hoogh et al., 2005; Jansen et al., 2009). The negative interaction of transactional leadership and innovation-oriented strategy indicates that a transactional leadership style may hinder, instead of foster, a company's implementation of an innovation-oriented strategy, in which organizational members truly live their innovation orientation. A transformational leadership style, which incorporates inspiration, motivation, and autonomy for organizational members, instead seems appropriate for implementing an innovation-oriented strategy for co-development projects. Because of its characteristics, transformational leadership contributes to advancing novel management practices, processes, and structures, by which organizational members become closely identified with the purpose and common goals of the organization (Vaccaro et al., 2012). The components of transformational leadership thus foster a company's implementation of an innovation-oriented strategy for co-development projects.

For the moderating effects of goods and services, a negative impact emerges on the relationship between innovation-oriented strategy and co-development, in line with arguments that service innovations can be rapidly implemented and copied and thus rarely are regarded as discretionary processes (Scarbrough and Lannon, 1989; Sundbo, 1997; Voss, 1992). Because the results for the moderating effects of goods and services indicate that senior managers' leadership does not vary in terms of effectiveness for co-development projects, only the implementation of an innovation-oriented strategy appears less supportive in service contexts.

In line with organizational learning theory, the empirical finding of a nonlinear relationship between co-development and new product frequency indicates that companies can profit from co-development at lower and medium levels, because information finds its way into the new product development process fairly easily. Organizational members do not appear overstrained by filtering and distributing the relevant information as needed. As co-development increases from medium to high levels though, customer information becomes more difficult to filter and interpret.

Organizational members must deal with more information, making it more challenging to identify useful, relevant information, such that the relationship turns negative (Figure 2).

In addition, this study offers an empirical test of the indirect effects of the senior management levers with new product frequency through co-development. When an innovation-oriented strategy is low, it fosters new product frequency through co-development. As this focus increases, no recognizable association appears with new product frequency through changes in co-development, whereas the indirect effect becomes negative for higher levels. That is, a highly innovation-oriented strategy may lower companies' new product frequency through co-development. With respect to the indirect effect of transformational leadership on new product frequency through co-development, increases at lower levels of transformational leadership are positively associated with new product frequency, but there is no significant indirect effect at moderate levels. If managers strongly pursue a very high level of transformational leadership, it even may reduce companies' new product frequency through co-development. Finally, transactional leadership reveals no indirect effect on new product frequency, consistent with the main model, in which the association of transactional leadership with co-development is not significant.

Theoretical Implications

From a theoretical perspective, this research relies on boundary theory and its wider contextualization within resource dependence theory to explicate senior management levers, as well as on organizational learning theory to explain the innovation outcomes of co-development. In line with resource dependence theory, senior management can serve as an important lever to bridge a company's boundary with customers in a co-development context. The application of organizational learning theory offers a route to explore the positive and negative innovation implications of co-development. Organizational learning theory is well established as a means to explain innovation generation and its outcomes, but resource dependence theory and boundary theory have rarely been applied in the context of co-development.

Whereas innovation-oriented strategy and transformational leadership provide more generalized overall guidance, transactional leadership offers relatively concrete

directions for the activities of organizational members (i.e., clear competencies, responsibilities, and expectations). These concrete directions are better suited to routines and difficult to transfer to more complex situations, such as co-development. The distinct consequences of the two leadership styles are further emphasized in light of their interactions with innovation-oriented strategy. Transformational leadership can be combined effectively with this strategy; transactional leadership in combination with an innovation-oriented strategy hinders co-development. Thus, this study adds a more nuanced view of the mechanisms by which strategy and leadership work together, as well as how they can be combined as complements.

This study also provides a more fine-grained understanding of the new product frequency outcomes of co-development. Drawing on organizational learning theory (Levitt and March, 1988; March, 1991), the authors argue for and find an inverted U-shaped relationship between co-development and new product frequency. Companies are less able to cope with very high levels of co-development, primarily because of the obligation to integrate and make use of newfound knowledge. Thus, too much co-development can harm innovativeness, and the notion of “the more, the better” does not apply to an unlimited degree.

Finally, this study contributes to the ongoing debate about whether goods and services should be treated differently by marketing professionals (Nijssen et al., 2006; Stock, 2011). Researchers increasingly refrain from this dichotomy, because goods and services both fulfill basic customer needs (Alam, 2006; Ali et al., 1995). The current results show that the link between an innovation-oriented strategy and co-development is negatively moderated by the goods or services variable, such that its positive impact on co-development is not as strong for services as it is for goods. For the associations between transformational and transactional leadership with co-development, no significant influence of the interactions between leadership styles and the goods or services variable arises. The impact of both leadership styles thus does not appear to differ with regard to whether goods or services are developed.

Managerial Implications

It is worthwhile for managerial practice to implement an innovation-oriented strategy, to successfully provide overall guidance in terms of a company's innovation generation goals. Senior managers' transformational leadership style ultimately can increase companies'

new product frequency through co-development, particularly in combination with an innovation-oriented strategy. Because the interaction of an innovation-oriented strategy and transformational leadership seems valuable for companies in terms of co-development projects, senior managers should adapt a transformational leadership style, whose expressed values, visions, and inspiration fit with an innovation-oriented strategy. Such senior managers can better motivate and lead organizational members, because the strategic direction of the company and the vision provided by senior managers are overlapping and consistent. Companies also should initiate appropriate training, coaching, and development to give senior managers opportunities to improve their ability to inspire their subordinates through intellectual stimulation, individualized consideration, and charisma (Ellam-Dyson and Palmer, 2011; Howell and Avolio, 1993). However, the empirical results indicate that the interaction between an innovation-oriented strategy and transactional leadership is negatively associated, so companies should avoid relying on fixed goals, expectations, and rewards, which hinder their flexible reactions to arising requirements in co-development projects.

Regarding the performance implications of co-development, managers can learn multiple lessons from its nonlinear relationship with new product frequency. First, co-development increases new product frequency by bringing new knowledge into companies, providing a basis for shorter new product development times. However, it is only effective and beneficial up to a certain degree. If companies focus too much on co-development, they encounter too much customer information, from which it is more complicated to extract useful and relevant information to incorporate into new product development projects (Hoyer et al., 2010). Second, companies should pursue a medium level of co-development, then carefully monitor and control that level. To discover when they have reached the limit, project management and innovation literature suggest various management tools and methods that senior managers can use to handle, plan, manage, and control their company's internal co-development processes to make them more effective and efficient (Bossink, 2002). For example, managers should track the progress of different projects using key performance indicators to recognize negative efficiency trends early. Customer ideas and input should be recorded, structured, and analyzed by managers, to determine when co-development efforts exceed their benefits. Managers

also should monitor the company's environment and target markets. By considering the requirements and needs of both customers involved in co-development and the rest of the market, managers can find support for their decisions to continue or stop a particular co-development project at a certain point in time.

These findings are relevant for managers in various contexts and industries. Transformational leadership, as a critical element of service co-development and an effective senior management lever for co-development, is required to increase new product frequency in a service context, in addition to innovation-oriented strategy. Companies offering services should attend to the leadership skills of their management staff, particularly in terms of transformational leadership. This finding affirms extant research that has identified leadership characteristics in general as a crucial factor for service creation (Aragón-Correa et al., 2007). Overall, in contrast with criticisms in strategic management literature (e.g., Gordon, Stewart, Sweo, and Luker, 2000), managers must learn that co-development can be fostered successfully using a top-down approach. Similar to other phenomena, co-development needs senior management support to thrive and cannot be implemented effectively on an operational level without it.

Limitations and Avenues for Further Research

Although this research provides multiple contributions, it is also subject to limitations that suggest avenues for further research. This study focuses on co-development as an activity for supplier companies to bridge the boundary with their customers. In this sense, it is limited to the intensity of co-development; the measures do not include actual boundary-spanning behaviors of the supplier or customers during co-development. In this context, especially customer-based data regarding their involvement in co-development would be desirable. This avenue is likely to provide interesting details regarding the operations of co-development activities.

With regard to the newly developed theoretical mechanism, this study did not include explicit measures of organizational openness according to aspects such as open communication, openness to experience, or openness to organizational change (Berson and Avolio, 2004; Groves, 2005; Moss, McFarland, Ngu, and Kijowska, 2007). Investigating this theoretical mechanism more deeply would be a valuable way for research to shed more light on the antecedents and

outcomes of co-development. Additional research also could measure new product frequency by the number of recently introduced products or examine other innovation outcomes, such as newness and meaningfulness.

The present analyses rely on multi-informant data from a cross-sectional survey. In this context, senior managers were asked to name the subordinates to participate in the study, which creates the possibility that the questionnaires were addressed only to certain categories of subordinates by senior managers, resulting in a potential bias due to endogeneity. Furthermore, with a longitudinal approach, researchers could investigate how changes in senior management levers affect co-development and subsequent new product frequency outcomes. In doing so, this stream of research could generate new knowledge about how to implement senior management levers for long-term co-development success.

Finally, this study's scope purposefully encompasses goods and services; many extant studies in the co-development context examine goods and services separately. By combining them, this research provides new insights about a boundary condition of co-development and shows that it is worthwhile to avoid an exclusive focus on either goods or services in further investigations. Researchers also might examine alternative contingency variables, such as customer or product-related characteristics that could influence the investigated relationships too.

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Appendix : Items and Scale Sources

Construct	Items	Factor Loadings
Innovation-Oriented Strategy^a (Stock and Zacharias, 2011); $\alpha = .80$; CR = .80; AVE = .56		
	Our company enters the market generally first with its products and services.	.74
	Our company does not hesitate to enter new market segments, which offer appropriate opportunities.	.78
	Our company intends to offer innovative products based on dramatic performance increases.	.73
Transformational Leadership^a (adapted from Bass and Avolio, 1995)		
	Intellectual Stimulation $\alpha = .73$; CR = .74; AVE = .50	
	As a leader, I ...	
	... seek differing perspectives when solving problems.	.81
	... get others to look at problems from many different angles.	.72
	... suggest new ways of looking at how to complete assignments.	.57
	Individualized Consideration $\alpha = .79$; CR = .80; AVE = .50	
	As a leader, I ...	
	... spend time teaching and coaching.	.60
	... treat others as individuals rather than just as a member of a group.	.75
	... consider the individual as having different needs, abilities, and aspirations from others.	.71
	... help others develop their strengths.	.75
	Charisma $\alpha = .82$; CR = .82; AVE = .70	
	As a leader, I ...	
	... am an inspiration to other people.	.82
	... make other people enthusiastic about assignments.	.85
Transactional Leadership^a (adapted from Bass and Avolio, 1995)		
	Contingent Rewards $\alpha = .85$; CR = .85; AVE = .65	
	As a leader, I ...	
	... make clear what subordinates receive, if their performance meets designated standards.	.88
	... work out agreements with subordinates what they will receive if they do what is to do.	.80
	... make sure that subordinates receive appropriate performance rewards for achieving performance targets.	.73
	Active Management by Exception $\alpha = .70$; CR = .74; AVE = .50	
	As a leader, I ...	
	... focus attention on irregularities, mistakes, exceptions, and deviations from standards.	.48
	... closely monitor subordinate's performance errors.	.95
	... enforce rules to avoid mistakes.	.61
Co-Development^a (adapted from Hsieh and Chen, 2005); $\alpha = .89$; CR = .89; AVE = .62		
	During the development of new products and services in our company ...	
	... we interact with customers beyond the standards of market research.	.71
	... customers participate in different stages of development projects.	.85
	... customers join our product development team, e.g., to test new products and services.	.85
	... we meet with customers to regard their opinion.	.71
	... we co-design products and services with customers.	.82
New Product Frequency^a (Stock and Zacharias, 2011); $\alpha = .93$; CR = .94; AVE = .75		
	My company ...	
	... has introduced more novel products/services during the last five years than our three strongest competitors.	.85
	... continually introduces innovative products/services into the market.	.94
	... replenishes or adds frequently novel products/services to its product offer.	.93
	... introduces many innovative products/services in the market.	.90
	... plans to introduce several innovations into the market during the next five years.	.69
Goods or Services^b		–
Number of Employees^c		–
Sales^d		–
	How high are the annual sales of your company (in million USD)?	–
Technology Turbulence^a (Jaworski and Kohli, 1993); $\alpha = .87$; CR = .87; AVE = .62		
	The technology in our industry is changing rapidly.	.83
	Technological changes provide big opportunities in our industry.	.85
	It is very difficult to forecast where the technology in our industry will be in the next 2 to 3 years.	.69
	A large number of new product ideas have been made possible through Technological breakthroughs in our industry.	.78

(Continued)

Appendix (Continued)**Competitive Intensity^a** (Jaworski and Kohli, 1993); $\alpha = .84$; CR = .84; AVE = .52

Competition in our industry is cutthroat.	.71
There are many “promotion wars” in our industry.	.83
Anything that one competitor can offer, others can match readily.	.67
Price competition is a hallmark of our industry.	.73
One hears of a new competitive move almost every day.	.65

Capital Intensity^a

The companies in the industry are capital intensive.	–
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Organizational Tenure^c

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Age of Senior Managers^c

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Senior Managers Position^f**Traditional Customer Information Acquisition^{a)}** (adapted from Jaworski and Kohli, 1993; Li and Calantone, 1998); $\alpha = .89$; CR = .89; AVE = .62

In our company ...	
... we regularly use external market research.	.81
... we regularly poll customers to identify their underlying needs.	.86
... we systematically process and analyze customer information.	.71
... we regularly use additional research procedures as personal interviews or focus groups to gather customer information.	.86
... employees regularly meet customers to learn about their needs.	.64

Notes: α = Cronbach's alpha; CR = composite reliability; AVE = average variance extracted.

^aItems measured with seven-point rating scales, with anchors at 1 = “strongly disagree” and 7 = “strongly agree.”

^bItem measured with a dummy-coded variable based on the industry sector information of the participating companies. Manufacturing, IT, machinery and electronics, and equipment suppliers were assigned to the goods category (0 = “goods”); professional services, banking, and retail represent services (1 = “services”).

^cItem measured with a 10-point scale, reflecting the number of employees, where 1 = <100 and 10 = >100,000.

^dItem measured with a 10-point scale, reflecting the annual sales of the company (millions USD), where 1 = <5 and 10 = >1000.

^eItem measured with a free-text field, in which respondents could indicate their organizational tenure in years.

^fItem measured with an effect-coded variable based on the position that senior managers held in companies (Senior Manager Position 1 = CEO; Senior Manager Position 2 = CIO, CFO, board member, upper management; Senior Manager Position 3 = others).